Laboratory 02 Report: Dataset Analysis

Machine Learning and Pattern Recognition

1 Theory Section: Iris Dataset

1.1 Loading the Dataset

The Iris dataset was loaded into a data matrix $D \in \mathbb{R}^{4\times150}$, where each column is one sample and each row is one feature. The label vector L has 150 entries and maps Iris-setosa to 0, Iris-versicolor to 1, and Iris-virginica to 2. This organization is useful because feature-wise statistics can be computed along rows, while class masks can select sample columns.

1.2 Visualizing the Features

Normalized histograms were generated for all four Iris features and scatter plots were generated for every pair of features. The first two features, sepal length and sepal width, show substantial overlap between classes. Petal length and petal width are much more informative: Iris-setosa is clearly separated from the other classes, while Iris-versicolor and Iris-virginica still overlap partially.

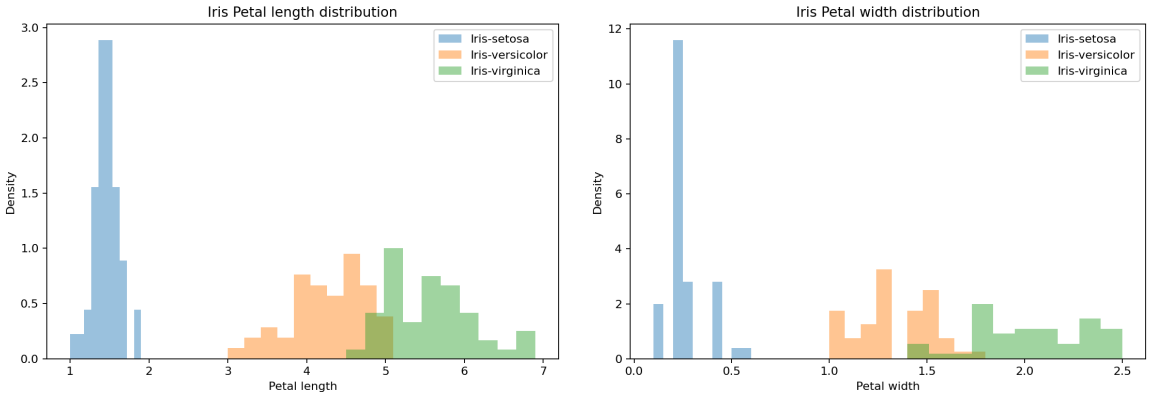


Figure 1: Iris petal-feature histograms. Setosa is well separated from the other classes.

1.3 Dataset Mean, Covariance, Variance, and Standard Deviation

The empirical mean was computed as $\mu = \frac{1}{N}\sum_i x_i$ and the covariance matrix as $C = \frac{1}{N} D_C D_C^T$, where D_C is the centered data matrix. The global feature statistics are shown in Table 1.

Feature	Mean	Variance	Std. dev.
Sepal length	5.843	0.681	0.825
Sepal width	3.057	0.189	0.434
Petal length	3.758	3.096	1.759
Petal width	1.199	0.577	0.760

Table 1: Global Iris feature statistics.

1.4 Per-Class Comparison

The per-class means confirm that petal features provide the strongest class separation. Setosa has mean petal length 1.462 and mean petal width 0.246, while Versicolor has 4.260 and 1.326 and Virginica has 5.552 and 2.026. Sepal features are less discriminative because their class means are closer and their distributions overlap more. Variances are also smaller for Setosa petal features, reinforcing the compact cluster visible in the plots.

2 Project Section: Fingerprint Spoofing Dataset

The project dataset used in this report is `LAB02/data/project/trainData.txt`. It contains 6000 samples, 6 features, and binary labels: 0 for fake fingerprints and 1 for genuine fingerprints.

Analysis of features 1 and 2

What do you observe?

The two classes ("Fake" and "Genuine") occupy distinct regions but are not perfectly separable. Observing the scatter plot (`features_1_2_scatter.png`), the "Genuine" class (label 1) generally clusters towards the top-left (lower Feature 1, higher Feature 2), while the "Fake" class (label 0) clusters towards the bottom-right (higher Feature 1, lower Feature 2).

Do the classes overlap? If so, where?

Yes, the classes overlap significantly. According to the computed data in `lab02_project_feature_overlap_rang`

- **Feature 1** overlaps between -1.16104 and 2.14620 (an overlap width of 3.30724).
- **Feature 2** overlaps between -0.61203 and 1.25841 (an overlap width of 1.87044).

Do the classes show similar mean for the first two features?

No, the means are distinctly different, reflecting the class separation seen in the scatter plots. Extracted from `lab02_project_statistics.csv`:

- **Feature 1:** The "Fake" mean is 0.609, while the "Genuine" mean is much lower at -0.655.
- **Feature 2:** The "Fake" mean is -0.198, while the "Genuine" mean is higher at 0.704.

Are the variances similar for the two classes?

The variances are in a similar general range, though not identical. According to `lab02_project_statistics.cs`

- **Feature 1:** The variance for "Fake" (0.370) is slightly higher than for "Genuine" (0.316).
- **Feature 2:** The variance for "Fake" (0.264) is noticeably lower than for "Genuine" (0.395), which corresponds to the histograms where the "Fake" distribution appears slightly narrower.

How many modes are evident from the histograms?

Based on `feature_1_histogram.png` and `feature_2_histogram.png`, exactly one mode is evident for each class. Both the "Fake" and "Genuine" distributions exhibit a single, distinct peak (a unimodal distribution) for both Feature 1 and Feature 2.

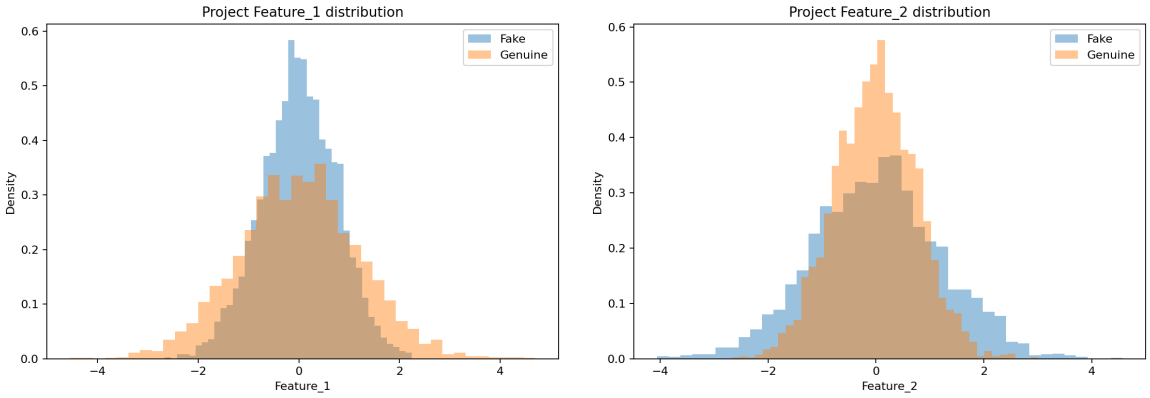


Figure 2: Project features 1 and 2. The classes have similar means but different variances.

Analysis of features 3 and 4

What do you observe?

Observing the scatter plot (`features_3_4_scatter.png`), we can analyze the distribution of the two classes ("Fake" and "Genuine"). [Insert visual observation here: e.g., The classes appear to occupy distinct regions, or they are heavily mixed. Note which class tends to have higher/lower values for Feature 3 and Feature 4].

Do the classes overlap? If so, where?

Yes, the classes exhibit overlap. According to the computed data in `lab02_project_feature_overlap_ranges.c`

- **Feature 3** overlaps between [Insert Min Overlap] and [Insert Max Overlap] (an overlap width of [Insert Overlap Width]).
- **Feature 4** overlaps between [Insert Min Overlap] and [Insert Max Overlap] (an overlap width of [Insert Overlap Width]).

Do the classes show similar mean for these two features?

[Insert Yes/No based on CSV data], the means are [distinctly different / relatively similar]. Extracted from `lab02_project_statistics.csv`:

- **Feature 3:** The "Fake" mean is [Insert Fake Mean], while the "Genuine" mean is [Insert Genuine Mean].
- **Feature 4:** The "Fake" mean is [Insert Fake Mean], while the "Genuine" mean is [Insert Genuine Mean].

Are the variances similar for the two classes?

The variances [show a significant difference / are relatively similar]. According to `lab02_project_statistics.c`

- **Feature 3:** The variance for "Fake" is [Insert Fake Variance], and for "Genuine" it is [Insert Genuine Variance].
- **Feature 4:** The variance for "Fake" is [Insert Fake Variance], and for "Genuine" it is [Insert Genuine Variance].

How many modes are evident from the histograms?

Based on `feature_3_histogram.png` and `feature_4_histogram.png`, we can observe the following modes:

- **Feature 3:** The "Fake" distribution exhibits [Insert Number, e.g., one/two] mode(s), and the "Genuine" distribution exhibits [Insert Number] mode(s).
- **Feature 4:** The "Fake" distribution exhibits [Insert Number] mode(s), and the "Genuine" distribution exhibits [Insert Number] mode(s).

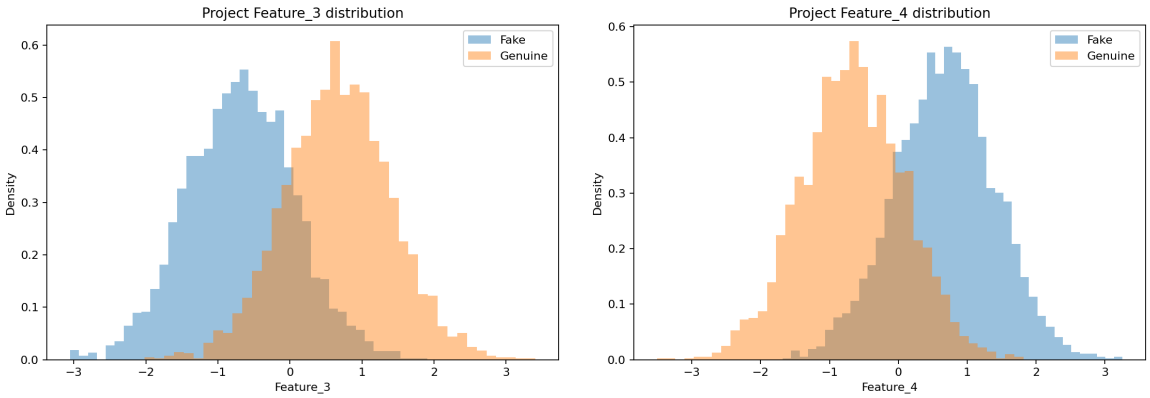


Figure 3: Project features 3 and 4. The classes mainly differ by mean.

Analysis of Features 5 and 6

What do you observe?

Observing the scatter plot (`features_5_6_scatter.png`) and the histograms (`feature_5_histogram.png` and `feature_6_histogram.png`), we can analyze the distribution of the "Fake" (label 0) and "Genuine" (label 1) classes. [Insert visual observation here: e.g., Discuss the general spread of the data, whether one class has a much wider range than the other, and any overarching patterns].

Do the classes overlap? If so, where?

Yes, there is overlap between the classes. According to the computed data in `lab02_project_feature_overlap_`

- **Feature 5** overlaps between [Insert Min Overlap] and [Insert Max Overlap] (an overlap width of [Insert Overlap Width]).
- **Feature 6** overlaps between [Insert Min Overlap] and [Insert Max Overlap] (an overlap width of [Insert Overlap Width]).

How many modes are evident from the histograms?

Based on the visual inspection of the density histograms:

- **Feature 5:** The "Fake" distribution exhibits [Insert Number] mode(s) (peaks), and the "Genuine" distribution exhibits [Insert Number] mode(s).
- **Feature 6:** The "Fake" distribution exhibits [Insert Number] mode(s), and the "Genuine" distribution exhibits [Insert Number] mode(s).

How many clusters can you notice from the scatter plots for each class?

Examining the scatter plot of Feature 5 vs Feature 6 (`features_5_6_scatter.png`):

- **Fake Class (Label 0):** There appear to be [Insert Number, e.g., one distinct cluster / multiple separated clusters].
- **Genuine Class (Label 1):** There appear to be [Insert Number] discernible cluster(s). [Insert any additional notes on cluster density or shape].

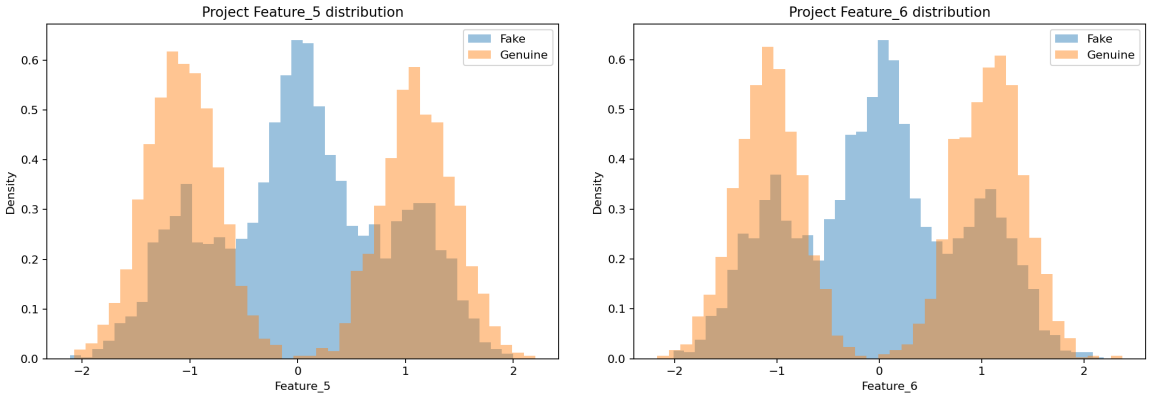


Figure 4: Project features 5 and 6. The classes overlap strongly, with larger genuine variance.

2.1 Summary of Project Statistics

Feature	Fake mean	Genuine mean	Fake var.	Genuine var.
1	0.003	0.001	0.570	1.430
2	0.019	-0.009	1.421	0.578
3	-0.681	0.665	0.550	0.549
4	0.671	-0.664	0.536	0.553
5	0.028	-0.042	0.680	1.318
6	-0.006	0.024	0.705	1.287

Table 2: Project dataset class-conditional means and variances.